

MODULE 9

GitHub Repository Link: <https://github.com/devcnx/csd440>

Deliverables

Drop Table — Clean slate before testing (or 'Nothing To Drop' if table absent)

Drop Table

Brittaney Perry-Morgan

Module 9.2 — Drop Table — CSD440 Server-Side Scripting

May 11, 2026

Table Dropped

The `bperymorgan_ollama_models` table was found and removed from the `baseball_01` database.

SQL executed: `DROP TABLE IF EXISTS bperymorgan_ollama_models`

Next step: rebuild the table by running [BrittaneyCreateTable.php](#).

[Index](#) [Create Table](#) [Populate Table](#) [Query Table](#)

Create Table — Schema confirmed with 8 columns

Create Table

Brittaney Perry-Morgan

Module 9.2 — Create Table — CSD440 Server-Side Scripting

May 11, 2026

Table Ready

The `bperrymorgan_ollama_models` table is in place (created if missing) inside the `baseball_01` database.

SQL executed: `CREATE TABLE IF NOT EXISTS bperrymorgan_ollama_models ( ... 8 columns ... )`

Next step: populate the table by running [BrittaneyPopulateTable.php](#).

Confirmed Schema

DESCRIBE bperrymorgan_ollama_models

Field	Type	Null	Key	Default	Extra
model_id	int	NO	PRI		auto_increment
model_name	varchar(100)	NO	UNI		
parameter_count	decimal(6,2)	NO			
quantization	varchar(20)	NO			
size_gb	decimal(6,2)	NO			
release_date	date	NO			
is_installed	tinyint(1)	NO		0	
use_case	varchar(255)	NO			

- Index
- Populate Table
- Query Table
- Drop Table

Populate Table — 8 seed rows loaded

Populate Table

Brittaney Perry-Morgan

Module 9.2 — Populate Table — CSD440 Server-Side Scripting

May 11, 2026

Seed Load Complete

Processed 8 rows against bperymorgan_ollama_models : 8 inserted, 0 updated.

SQL pattern: INSERT INTO bperymorgan_ollama_models (_) VALUES (?, ?, ?, ?, ?, ?, ?) ON DUPLICATE KEY UPDATE _ — bound seven times per row using mysql::prepare + bind_param .

Next step: verify with [BrittaneyQueryTable.php](#) .

Table Contents After Load

SELECT * FROM bperymorgan_ollama_models ORDER BY model_id

ID	Model	Params (B)	Quant	Size (GB)	Released	Installed	Use Case
1	llava:13b	13.00	Q4_K_M	8.00	2024-01-30	No	Vision-language model for image understanding
2	nomic-embed-text:latest	0.14	F16	0.27	2024-02-01	Yes	Text embeddings for RAG pipelines
3	phi3:mini	3.80	Q4_K_M	2.30	2024-04-23	Yes	Lightweight reasoning on limited hardware
4	gemma2:9b	9.00	Q4_K_M	5.44	2024-06-27	Yes	Open Google model, balanced performance
5	mistral-nemo:12b	12.00	Q4_K_M	7.07	2024-07-18	Yes	Multilingual chat with long context window
6	llama3.1:8b	8.00	Q4_K_M	4.92	2024-07-23	Yes	General-purpose chat and coding assistance
7	llama3.1:70b	70.00	Q4_K_M	39.50	2024-07-23	No	High-quality reasoning, slower local inference
8	qwen2.5-coder:7b	7.00	Q4_K_M	4.36	2024-09-19	Yes	Code generation and refactoring

Query Table — Four static SELECT queries against populated data

Query Table

Brittaney Perry-Morgan

Module 9.2 — Query Table — CSD440 Server-Side Scripting

May 11, 2026

1. All Models (newest first)

SQL: SELECT * FROM bperymorgan_ollama_models ORDER BY release_date DESC, model_name

All 8 models

ID	Model	Params (B)	Quant	Size (GB)	Released	Installed	Use Case
8	qwen2.5-coder:7b	7.00	Q4_K_M	4.36	2024-09-19	Yes	Code generation and refactoring
7	llama3.1:70b	70.00	Q4_K_M	39.50	2024-07-23	No	High-quality reasoning, slower local inference
6	llama3.1:8b	8.00	Q4_K_M	4.92	2024-07-23	Yes	General-purpose chat and coding assistance
5	mistral-nemo:12b	12.00	Q4_K_M	7.07	2024-07-18	Yes	Multilingual chat with long context window
4	gemma2:9b	9.00	Q4_K_M	5.44	2024-06-27	Yes	Open Google model, balanced performance
3	phi3:mini	3.80	Q4_K_M	2.30	2024-04-23	Yes	Lightweight reasoning on limited hardware
2	nomic-embed-text:latest	0.14	F16	0.27	2024-02-01	Yes	Text embeddings for RAG pipelines
1	llava:13b	13.00	Q4_K_M	8.00	2024-01-30	No	Vision-language model for image understanding

2. Locally Installed Models, Largest First

SQL: SELECT model_name, parameter_count, size_gb, use_case FROM bperymorgan_ollama_models WHERE is_installed = 1 ORDER BY parameter_count DESC

6 installed models

Model	Params (B)	Size (GB)	Use Case
mistral-nemo:12b	12.00	7.07	Multilingual chat with long context window
gemma2:9b	9.00	5.44	Open Google model, balanced performance
llama3.1:8b	8.00	4.92	General-purpose chat and coding assistance
qwen2.5-coder:7b	7.00	4.36	Code generation and refactoring
phi3:mini	3.80	2.30	Lightweight reasoning on limited hardware
nomic-embed-text:latest	0.14	0.27	Text embeddings for RAG pipelines

3. Fleet Summary (aggregates)

SQL: SELECT COUNT(*), SUM(is_installed), ROUND(SUM(size_gb),2), ROUND(AVG(parameter_count),2), MAX(parameter_count), MIN(release_date), MAX(release_date) FROM bperymorgan_ollama_models

Aggregate Statistics

Total models cataloged	8
Installed locally	6
Combined disk footprint	71.86 GB
Average parameter count	15.37 B
Largest model	70.00 B parameters
Oldest release date	2024-01-30
Newest release date	2024-09-19

4. Models Grouped by Quantization

SQL: SELECT quantization, COUNT(*), ROUND(AVG(size_gb),2) FROM bperymorgan_ollama_models GROUP BY quantization ORDER BY model_count DESC

Counts per quantization scheme

Quantization	Model Count	Avg Size (GB)
Q4_K_M	7	10.23
F16	1	0.27

Index

Create Table

Populate Table

Drop Table

Index — Database status showing 8 rows, navigation to all pages

Ollama Models Manager

Brittaney Perry-Morgan

Module 9.2 Programming Assignment — Index — CSD440 Server-Side Scripting

May 11, 2026

Database Status

Table `bperrymorgan_ollama_models` is present (8 rows).

Module 9 — New Pages

Page	Description
Index	This page — site map and database status check
Query	Search models by name, quantization, installed status, or parameter range
Add Record	Insert a new Ollama model into the database via a validated form

Module 8 — Table Management

Page	Description
Create Table	Creates the <code>bperrymorgan_ollama_models</code> table (idempotent)
Populate Table	Loads the 8-row seed dataset (INSERT ... ON DUPLICATE KEY UPDATE)
Query Table	Four predefined SELECT queries against the table
Drop Table	Removes the table for a clean reset (DROP TABLE IF EXISTS)

May 11, 2026

Add Record — Validation errors when submitted empty

Add a Model Record

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Module 9.2 Programming Assignment — Add Record — CSD440 Server-Side Scripting

May 11, 2026

Model Name *

e.g., llama3.1:8b

Parameter Count (Billions) *

e.g., 8.00

Quantization *

e.g., Q4_K_M

Size (GB) *

e.g., 4.92

Release Date *

mm/dd/yyyy

Installed Locally *

— Select —

Use Case *

e.g., General-purpose chat and coding assistance

Add Record

Reset

* Required fields

Index

Search Models

Add Record — Successful insertion of test-model:7b

Add a Model Record

Brittaney Perry-Morgan

Module 9.2 Programming Assignment — Add Record — CSD440 Server-Side Scripting

May 11, 2026

Record Added Successfully

The model has been inserted into the `bperrymorgan_ollama_models` table.

Inserted Record

Model Name	test-model:7b
Parameter Count	7.00 B
Quantization	Q4_K_M
Size (GB)	4.50 GB
Release Date	2025-01-15
Installed Locally	No
Use Case	Test model for module 9

Add Another Record

Search Models

Index

Index

Search Models

Search — All models with no filters applied (9 rows)

Search Models

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Module 9.2 Programming Assignment — Query — CSD440 Server-Side Scripting

May 11, 2026

Model Name

Partial match (e.g., llama)

Quantization

— Any —

Installed Locally

— Any —

Parameter Count (Min, billions)

e.g., 1

Parameter Count (Max, billions)

e.g., 100

Search

Reset

Enter search criteria above and click **Search** to filter models. Leave all fields blank to list every row.

Index

Add Record

Search — Filtering by installed status (Yes)

Search Models

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Module 9.2 Programming Assignment — Query — CSD440 Server-Side Scripting

May 11, 2026

Model Name

Partial match (e.g., llama)

Quantization

— Any —

Installed Locally

Yes

Parameter Count (Min, billions)

e.g., 1

Parameter Count (Max, billions)

e.g., 100

Search

Reset

Search Results

Filters applied: Model name: any, Quantization: any, Installed: Yes, Params: — — —

6 models found

ID	Model	Params (B)	Quant	Size (GB)	Released	Installed	Use Case
2	nomic-embed-text:latest	0.14	F16	0.27	2024-02-01	Yes	Text embeddings for RAG pipelines
3	phi3:mini	3.80	Q4_K_M	2.30	2024-04-23	Yes	Lightweight reasoning on limited hardware
8	qwen2.5-coder:7b	7.00	Q4_K_M	4.36	2024-09-19	Yes	Code generation and refactoring
6	llama3.1:8b	8.00	Q4_K_M	4.92	2024-07-23	Yes	General-purpose chat and coding assistance
4	gemma2:9b	9.00	Q4_K_M	5.44	2024-06-27	Yes	Open Google model, balanced performance
5	mistral-nemo:12b	12.00	Q4_K_M	7.07	2024-07-18	Yes	Multilingual chat with long context window

Search — Parameter range filter (10–20 billion)

Search Models

Brittany Perry-Morgan

Module 9.2 Programming Assignment — Query — CSD440 Server-Side Scripting

May 11, 2026

Model Name

Quantization

— Any —

Installed Locally

— Any —

Parameter Count (Min, billions)

Parameter Count (Max, billions)

Search

Reset

Search Results

Filters applied: Model name: any, Quantization: any, Installed: any, Params: 10B – 20B

2 models found

ID	Model	Params (B)	Quant	Size (GB)	Released	Installed	Use Case
5	mistral-nemo:12b	12.00	Q4_K_M	7.07	2024-07-18	Yes	Multilingual chat with long context window
1	llava:13b	13.00	Q4_K_M	8.00	2024-01-30	No	Vision-language model for image understanding

Index

Add Record

Search — No results for 'zzzzz'

Search Models

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Module 9.2 Programming Assignment — Query — CSD440 Server-Side Scripting

May 11, 2026

Model Name

zzzzz

Quantization

— Any —

Installed Locally

— Any —

Parameter Count (Min, billions)

e.g., 1

Parameter Count (Max, billions)

e.g., 100

Search

Reset

Search Results

Filters applied: Model name: "zzzzz", Quantization: any, Installed: any, Params: — — —

No models matched your search criteria. Try broadening the filters.

Index

Add Record

Query Table — Shows 9 rows after adding test record

Query Table

Brittaney Perry-Morgan

Module 9.2 — Query Table — CSD440 Server-Side Scripting

May 11, 2026

1. All Models (newest first)

SQL: SELECT * FROM bperrymorgan_ollama_models ORDER BY release_date DESC, model_name

All 9 models

ID	Model	Params (B)	Quant	Size (GB)	Released	Installed	Use Case
9	test-model:7b	7.00	Q4_K_M	4.50	2025-01-15	No	Test model for module 9
8	qwen2.5-coder:7b	7.00	Q4_K_M	4.36	2024-09-19	Yes	Code generation and refactoring
7	llama3.1:70b	70.00	Q4_K_M	39.50	2024-07-23	No	High-quality reasoning, slower local inference
6	llama3.1:8b	8.00	Q4_K_M	4.92	2024-07-23	Yes	General-purpose chat and coding assistance
5	mistral-nemo:12b	12.00	Q4_K_M	7.07	2024-07-18	Yes	Multilingual chat with long context window
4	gemma2:9b	9.00	Q4_K_M	5.44	2024-06-27	Yes	Open Google model, balanced performance
3	phi3:mini	3.80	Q4_K_M	2.30	2024-04-23	Yes	Lightweight reasoning on limited hardware
2	nonic-embed-text:latest	0.14	F16	0.27	2024-02-01	Yes	Text embeddings for RAG pipelines
1	llava:13b	13.00	Q4_K_M	8.00	2024-01-30	No	Vision-language model for image understanding

2. Locally Installed Models, Largest First

SQL: SELECT model_name, parameter_count, size_gb, use_case FROM bperrymorgan_ollama_models WHERE is_installed = 1 ORDER BY parameter_count DESC

6 installed models

Model	Params (B)	Size (GB)	Use Case
mistral-nemo:12b	12.00	7.07	Multilingual chat with long context window
gemma2:9b	9.00	5.44	Open Google model, balanced performance
llama3.1:8b	8.00	4.92	General-purpose chat and coding assistance
qwen2.5-coder:7b	7.00	4.36	Code generation and refactoring
phi3:mini	3.80	2.30	Lightweight reasoning on limited hardware
nonic-embed-text:latest	0.14	0.27	Text embeddings for RAG pipelines

3. Fleet Summary (aggregates)

SQL: SELECT COUNT(*), SUM(is_installed), ROUND(SUM(size_gb),2), ROUND(AVG(parameter_count),2), MAX(parameter_count), MIN(release_date), MAX(release_date) FROM bperrymorgan_ollama_models

Aggregate Statistics

Total models cataloged	9
Installed locally	6
Combined disk footprint	76.36 GB
Average parameter count	14.44 B
Largest model	70.00 B parameters
Oldest release date	2024-01-30
Newest release date	2025-01-15

4. Models Grouped by Quantization

SQL: SELECT quantization, COUNT(*), ROUND(AVG(size_gb),2) FROM bperrymorgan_ollama_models GROUP BY quantization ORDER BY model_count DESC

Counts per quantization scheme

Quantization	Model Count	Avg Size (GB)
Q4_K_M	8	9.51
F16	1	0.27

Index

Create Table

Populate Table

Drop Table

Drop Table — Table dropped successfully

Drop Table

Brittaney Perry-Morgan

Module 9.2 — Drop Table — CSD440 Server-Side Scripting

May 11, 2026

Table Dropped

The `bperymorgan_ollama_models` table was found and removed from the `baseball_01` database.

SQL executed: `DROP TABLE IF EXISTS bperymorgan_ollama_models`

Next step: rebuild the table by running [BrittaneyCreateTable.php](#).

[Index](#) [Create Table](#) [Populate Table](#) [Query Table](#)

Drop Table — Idempotent re-run ('Nothing To Drop')

Drop Table

Brittaney Perry-Morgan

Module 9.2 — Drop Table — CSD440 Server-Side Scripting

May 11, 2026

Nothing To Drop

The `bperymorgan_ollama_models` table did not exist, so the DROP statement completed as a no-op. Database is in a clean state.

SQL executed: `DROP TABLE IF EXISTS bperymorgan_ollama_models`

Next step: rebuild the table by running [BrittaneyCreateTable.php](#).

[Index](#) [Create Table](#) [Populate Table](#) [Query Table](#)

Query Table — Errors after table has been dropped

Query Table

Brittaney Perry-Morgan

Module 9.2 — Query Table — CSD440 Server-Side Scripting

May 11, 2026

1. All Models (newest first)

SQL: SELECT * FROM bperrymorgan_ollama_models ORDER BY release_date DESC, model_name

Query failed: Table 'baseball_01.bperrymorgan_ollama_models' doesn't exist

2. Locally Installed Models, Largest First

SQL: SELECT model_name, parameter_count, size_gb, use_case FROM bperrymorgan_ollama_models WHERE is_installed = 1 ORDER BY parameter_count DESC

Query failed: Table 'baseball_01.bperrymorgan_ollama_models' doesn't exist

3. Fleet Summary (aggregates)

SQL: SELECT COUNT(*), SUM(is_installed), ROUND(SUM(size_gb),2), ROUND(AVG(parameter_count),2), MAX(parameter_count), MIN(release_date), MAX(release_date) FROM bperrymorgan_ollama_models

Query failed: Table 'baseball_01.bperrymorgan_ollama_models' doesn't exist

4. Models Grouped by Quantization

SQL: SELECT quantization, COUNT(*), ROUND(AVG(size_gb),2) FROM bperrymorgan_ollama_models GROUP BY quantization ORDER BY model_count DESC

Query failed: Table 'baseball_01.bperrymorgan_ollama_models' doesn't exist

Index

Create Table

Populate Table

Drop Table

Add Record — INSERT fails without table

Add a Model Record

Brittaney Perry-Morgan

Module 9.2 Programming Assignment — Add Record — CSD440 Server-Side Scripting

May 11, 2026

Correction Required

Please fix the following issues and resubmit:

- Table 'baseball_01.bperrymorgan_ollama_models' doesn't exist

Model Name *

orphan-model

Parameter Count (Billions) *

1

Quantization *

F16

Size (GB) *

0.5

Release Date *

06/01/2025

Installed Locally *

Yes

Use Case *

test

Add Record

Reset

* Required fields

Index

Search Models

Create Table — Re-created after drop

Create Table

Brittaney Perry-Morgan

Module 9.2 — Create Table — CSD440 Server-Side Scripting

May 11, 2026

Table Ready

The `bperrymorgan_ollama_models` table is in place (created if missing) inside the `baseball_01` database.

SQL executed: `CREATE TABLE IF NOT EXISTS bperrymorgan_ollama_models ( ... 8 columns ... )`

Next step: populate the table by running [BrittaneyPopulateTable.php](#).

Confirmed Schema

DESCRIBE bperrymorgan_ollama_models

Field	Type	Null	Key	Default	Extra
model_id	int	NO	PRI		auto_increment
model_name	varchar(100)	NO	UNI		
parameter_count	decimal(6,2)	NO			
quantization	varchar(20)	NO			
size_gb	decimal(6,2)	NO			
release_date	date	NO			
is_installed	tinyint(1)	NO		0	
use_case	varchar(255)	NO			

- Index
- Populate Table
- Query Table
- Drop Table

Populate Table — Re-seeded with 8 rows

Populate Table

Brittaney Perry-Morgan

Module 9.2 — Populate Table — CSD440 Server-Side Scripting

May 11, 2026

Seed Load Complete

Processed 8 rows against `bperrymorgan_ollama_models` : 8 inserted, 0 updated.

SQL pattern: `INSERT INTO bperrymorgan_ollama_models ( _ ) VALUES ( ?, ?, ?, ?, ?, ?, ? ) ON DUPLICATE KEY UPDATE _` — bound seven times per row using `mysql::prepare + bind_param`.

Next step: verify with [BrittaneyQueryTable.php](#).

Table Contents After Load

SELECT * FROM bperrymorgan_ollama_models ORDER BY model_id

ID	Model	Params (B)	Quant	Size (GB)	Released	Installed	Use Case
1	llava:13b	13.00	Q4_K_M	8.00	2024-01-30	No	Vision-language model for image understanding
2	nomic-embed-text:latest	0.14	F16	0.27	2024-02-01	Yes	Text embeddings for RAG pipelines
3	phi3:mini	3.80	Q4_K_M	2.30	2024-04-23	Yes	Lightweight reasoning on limited hardware
4	gemma2:9b	9.00	Q4_K_M	5.44	2024-06-27	Yes	Open Google model, balanced performance
5	mistral-nemo:12b	12.00	Q4_K_M	7.07	2024-07-18	Yes	Multilingual chat with long context window
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7	llama3.1:70b	70.00	Q4_K_M	39.50	2024-07-23	No	High-quality reasoning, slower local inference
8	qwen2.5-coder:7b	7.00	Q4_K_M	4.36	2024-09-19	Yes	Code generation and refactoring

Index — Final state confirming 8 rows

Ollama Models Manager

Brittaney Perry-Morgan

Module 9.2 Programming Assignment — Index — CSD440 Server-Side Scripting

May 11, 2026

Database Status

Table `bperrymorgan_ollama_models` is present (8 rows).

Module 9 — New Pages

Page	Description
Index	This page — site map and database status check
Query	Search models by name, quantization, installed status, or parameter range
Add Record	Insert a new Ollama model into the database via a validated form

Module 8 — Table Management

Page	Description
Create Table	Creates the <code>bperrymorgan_ollama_models</code> table (idempotent)
Populate Table	Loads the 8-row seed dataset (INSERT ... ON DUPLICATE KEY UPDATE)
Query Table	Four predefined SELECT queries against the table
Drop Table	Removes the table for a clean reset (DROP TABLE IF EXISTS)